

Data Science Capstone Project Report

Predicting SpaceX Falcon 9 First-Stage Landing Success

Your Name | October 2026

GitHub: <https://github.com/YOUR-USERNAME/spacex-falcon9-capstone>

Figures use a simulated dataset shaped like the real Falcon 9 data; rerun the repo on API data for real results.

Executive Summary

- Goal: predict whether a Falcon 9 first stage lands, which drives launch cost (~\$62M vs ~\$165M).
- Methods: SpaceX REST API + Wikipedia scraping, pandas wrangling, EDA (matplotlib), SQL (sqlite),
 - Folium maps, Plotly Dash dashboard, and ML (LogReg, SVM, Decision Tree, KNN with 10-fold GridSearchCV).
- Key results: landing success rises from 7% (flights 1-30) to 93% (flights 61+).
- Flight number (accumulated experience) is the clearest driver of success in this dataset; orbit, payload and site are secondary.
- Best model on the test set: Decision Tree at 89% accuracy (all four models within a narrow band).
- Takeaway: a competitor can estimate Falcon 9 launch cost by forecasting booster reuse probability.

Introduction

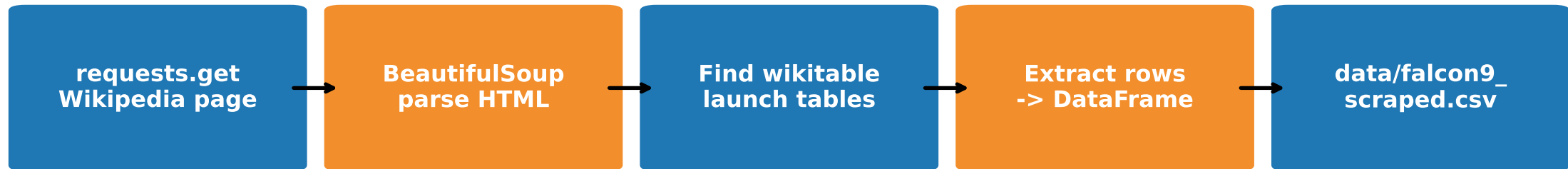
- Background: Falcon 9 launches cost ~\$62M vs ~\$165M+ for others, because Stage 1 is reusable.
- If we can predict whether Stage 1 will land, we can predict launch cost.
- Problem statement: given launch features (payload, orbit, site, flight number, reuse, grid fins, legs),
 - classify each launch as landing success (1) or failure (0).
- Questions: Which factors drive success? Do sites differ? How did success improve over time?
- Audience: a new rocket company bidding against SpaceX.

Data Collection - SpaceX API



Key phrases: `requests.get()` | `response.json()` | `pd.json_normalize()` | ID look-ups cached | drop Falcon 1 | save CSV

Data Collection - Web Scraping



Key phrases: soup.find_all('table') | row cells -> text | booster landing outcome | payload mass | orbit | customer

Data Wrangling Methodology

- Merged API and scraped tables on flight number; kept Falcon 9 launches only.
- Imputed missing PayloadMass with the column mean; kept LandingPad nulls (= no pad used).
- Counted launches per site and orbit, and landing outcomes (True/False Ocean, RTLS, ASDS, None).
- Created binary label Class: 1 = successful landing, 0 = otherwise.
- Result: 90 rows, overall success rate 50%.
- One-hot encoded Orbit and LaunchSite, standardised features before modelling.

EDA with Data Visualization

- Scatter: FlightNumber vs LaunchSite (hue = outcome) - does experience help at each site?
- Scatter: PayloadMass vs LaunchSite - does payload weight limit success?
- Bar: success rate by orbit - which missions are safest?
- Scatter: FlightNumber vs Orbit - orbit-specific learning curve.
- Line: yearly success-rate trend - overall improvement.
- Purpose: spot patterns that guide feature selection for the classifier.

EDA with SQL

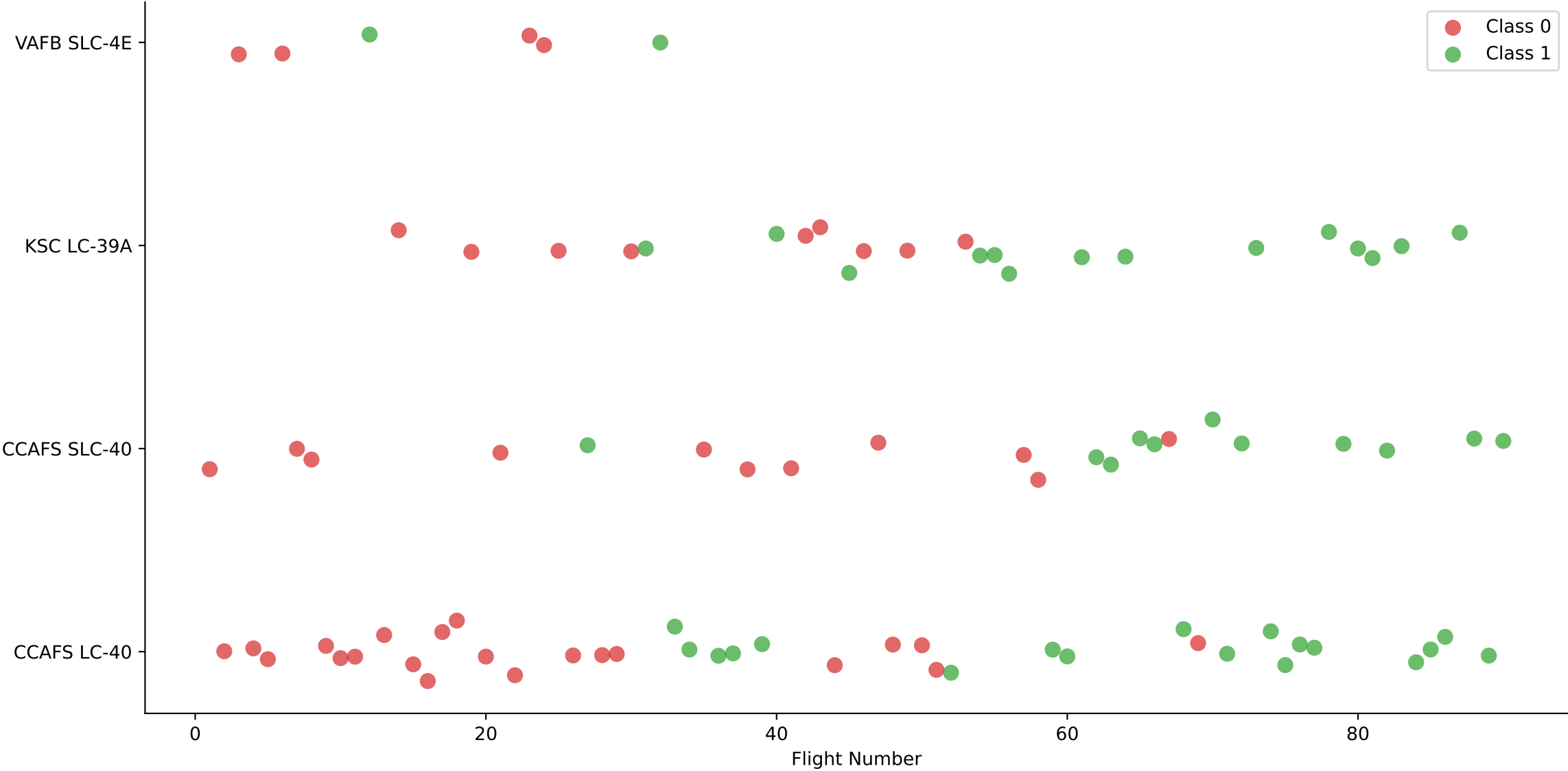
- Loaded data into SQLite (SPACEXTBL) and ran 9 queries.
- DISTINCT launch sites; LIKE 'CCA%' filter with LIMIT 5.
- SUM / AVG of payload by orbit and for reused boosters.
- MIN(date) of first successful landing; MAX payload landed.
- GROUP BY outcome counts, landings per year, success rate per site.
- ORDER BY ... DESC to rank years and sites.

Interactive Visual Analytics

- Folium: markers for 4 launch sites, green/red outcome markers per launch (MarkerCluster).
- Folium: MousePosition + PolyLine to measure distance to coast, railway, highway and city.
- Plotly Dash: dropdown for site + RangeSlider for payload.
- Dash: success pie chart (all sites or one site) and payload-vs-outcome scatter by booster version.
- Dash app: https://github.com/YOUR-USERNAME/spacex-falcon9-capstone/blob/main/src/dash_app.py | Map: https://github.com/YOUR-USERNAME/spacex-falcon9-capstone/blob/main/src/folium_map.py

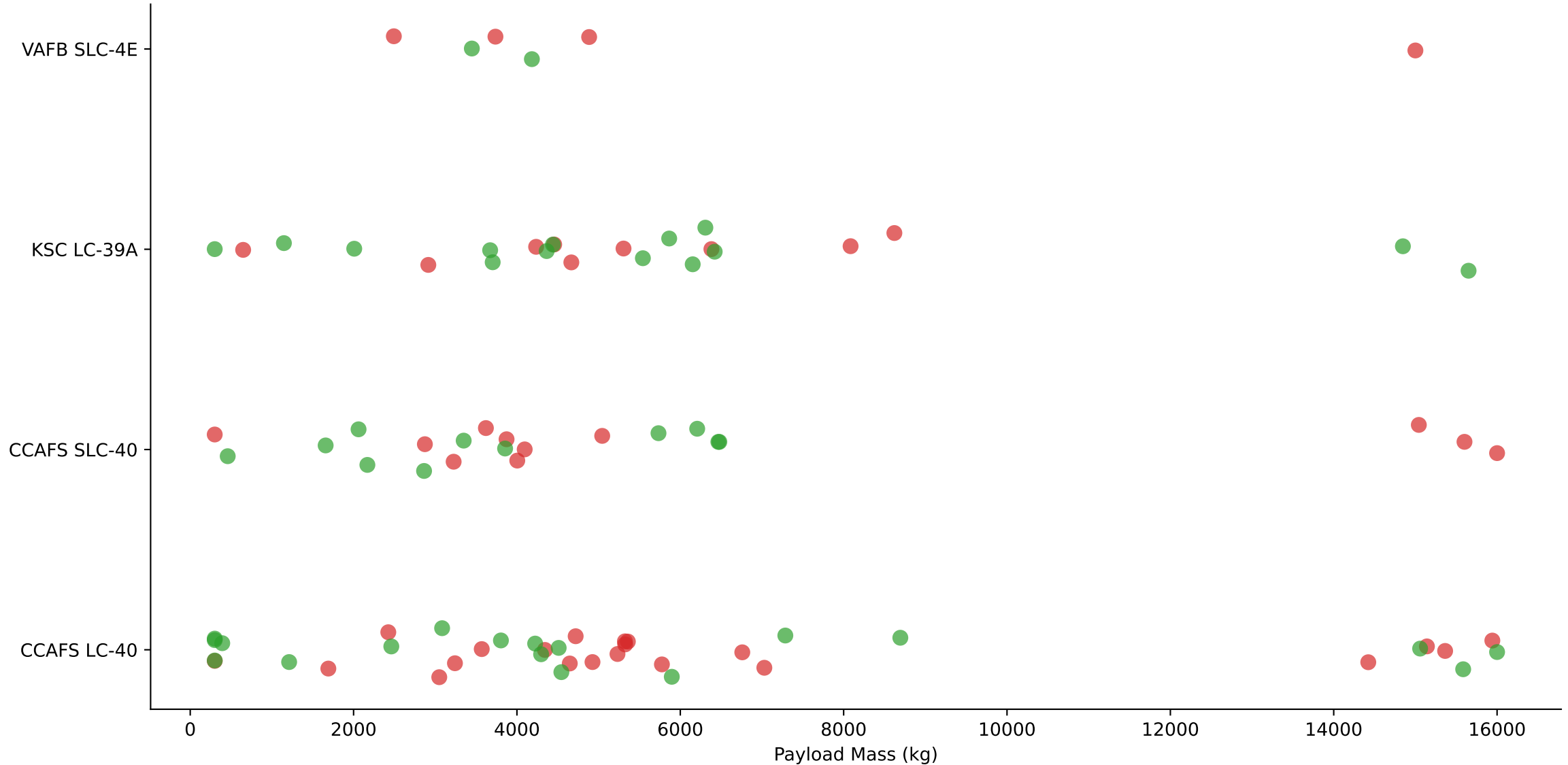
EDA: Flight Number vs Launch Site

Success clusters at higher flight numbers at every site: experience improves outcomes.



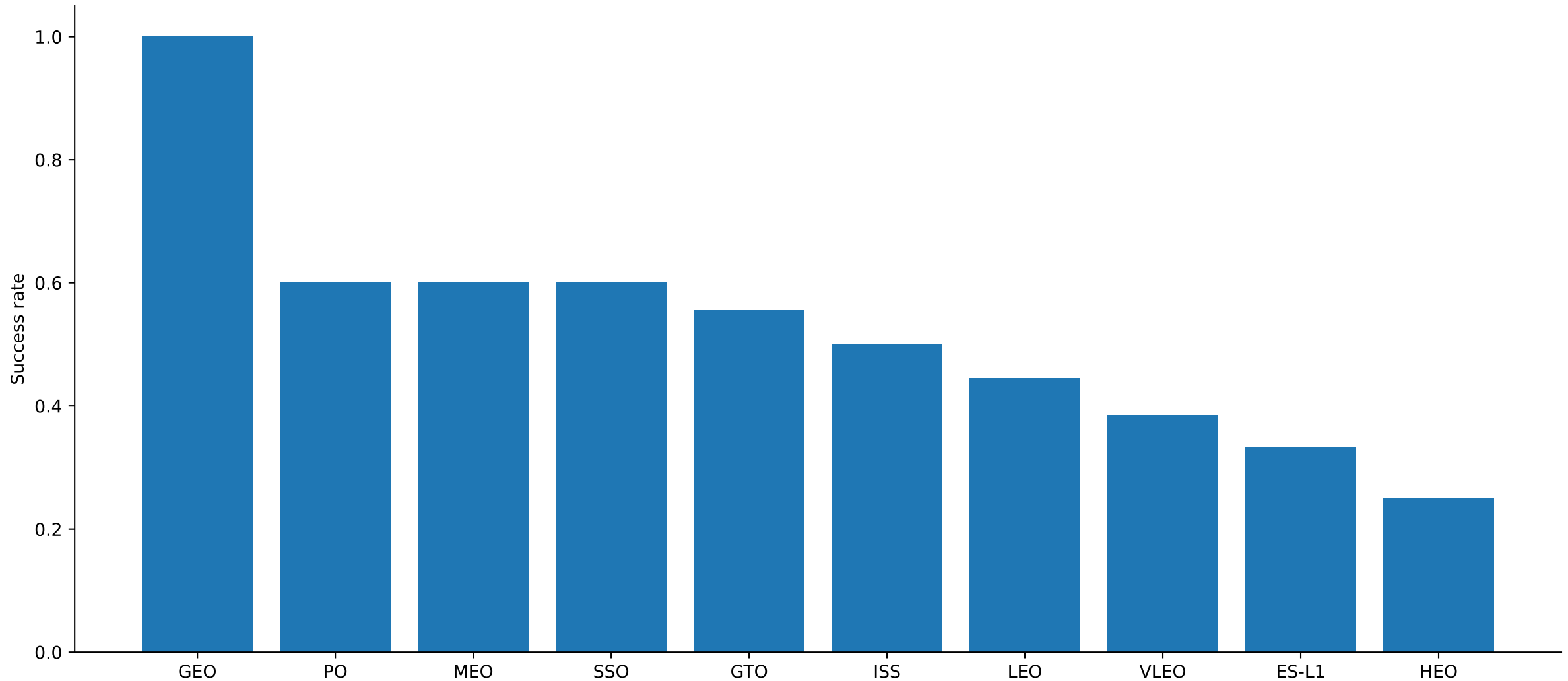
EDA: Payload vs Launch Site

Payloads cluster at 2-7 t; a separate group above 14 t (VLEO). Heavy-payload success rate: 38%.

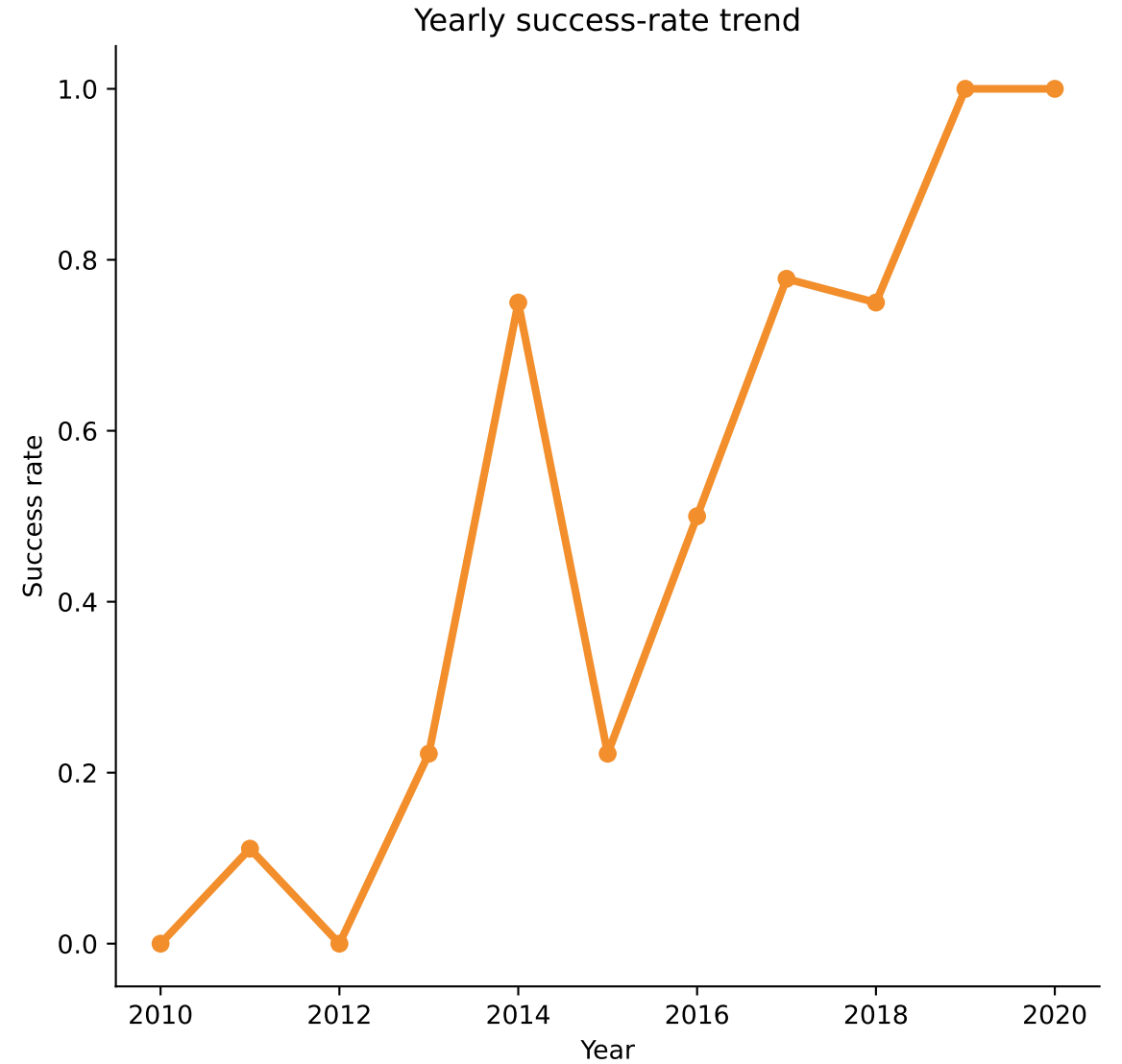
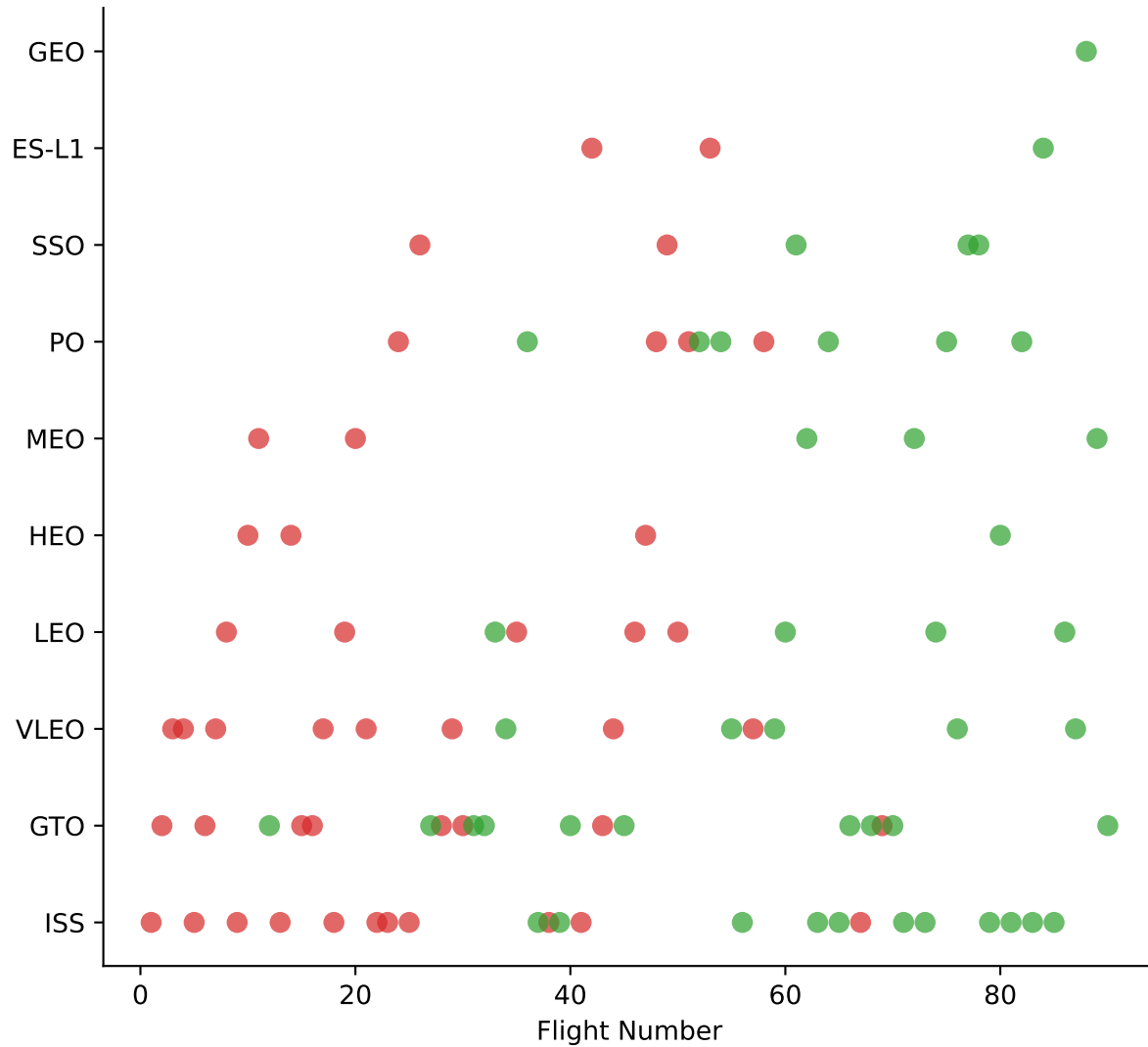


EDA: Success Rate by Orbit

Highest: GEO (100%); lowest: HEO (25%). Small orbit groups ($n \leq 5$) make these rates noisy.



EDA: Flight Number vs Orbit and Yearly Trend



EDA with SQL: Sites, Payloads, Dates

Unique launch sites

LaunchSite
CCAFS SLC-40
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A

5 records where site starts with 'CCA'

FlightNumber	LaunchSite	Orbit	PayloadMass
1	CCAFS SLC-40	ISS	4095.0
2	CCAFS LC-40	GTO	5322.0
4	CCAFS LC-40	VLEO	14423.0
5	CCAFS LC-40	ISS	4343.0
7	CCAFS SLC-40	VLEO	16000.0

Aggregate queries

Query	Result
Total payload, ISS (kg)	92614.0
Avg payload, reused (kg)	5036.0
First successful landing	2011-09-20
Heaviest payload landed (kg)	16000.0

EDA with SQL: Success Rates, Rankings, Time

Outcome counts (Class)

Class	Count
0	45
1	45

Top years by landings

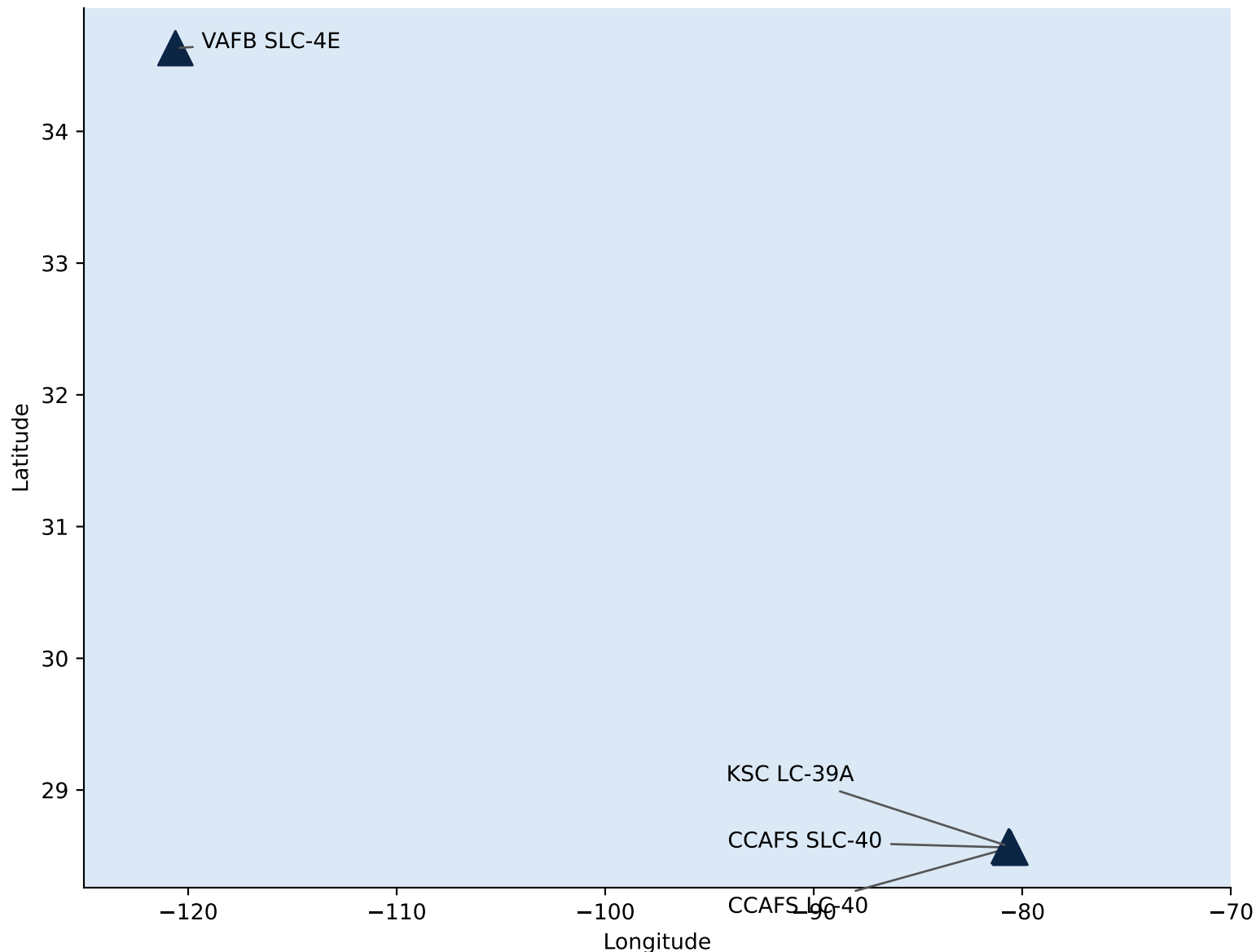
Year	landings
2019	9
2020	8
2017	7
2018	6
2014	6
2016	4
2015	2

Success rate by site (ranked)

LaunchSite	rate
KSC LC-39A	0.61
CCAFS SLC-40	0.5
CCAFS LC-40	0.46
VAFB SLC-4E	0.33

Folium Map: Launch Sites and Launch Records

Launch sites (schematic of Folium markers)



- 3 of 4 sites are in Florida;
- 1 (VAFB) is in California.
- All sites sit close to the coast
- (launches go over ocean).
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- Cluster markers: green = success,
- red = failure; count per site
- shown on cluster badge.
- Best site by rate: KSC LC-39A

Folium Map: Proximity Analysis

KSC LC-39A (approx., measured with MousePosition + PolyLine)

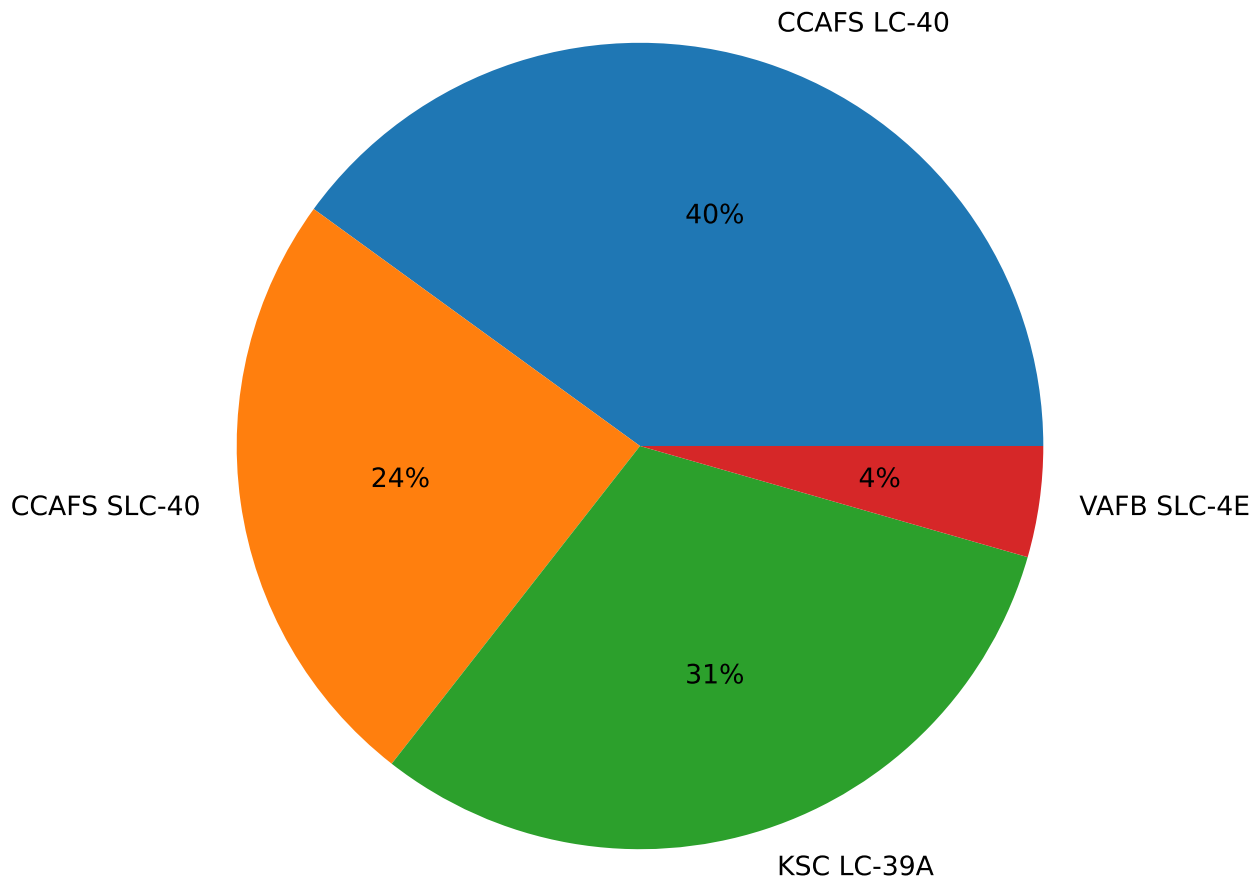
Feature	Distance (km)
Coastline	0.9
Railway	1.3
Highway	0.6
Nearest city (Titusville)	23.0

- Near coastline: yes - rockets fly over water to limit risk.
- Near railway/highway: yes - heavy equipment transport.
- Far from cities: yes - safety buffer from populated areas.

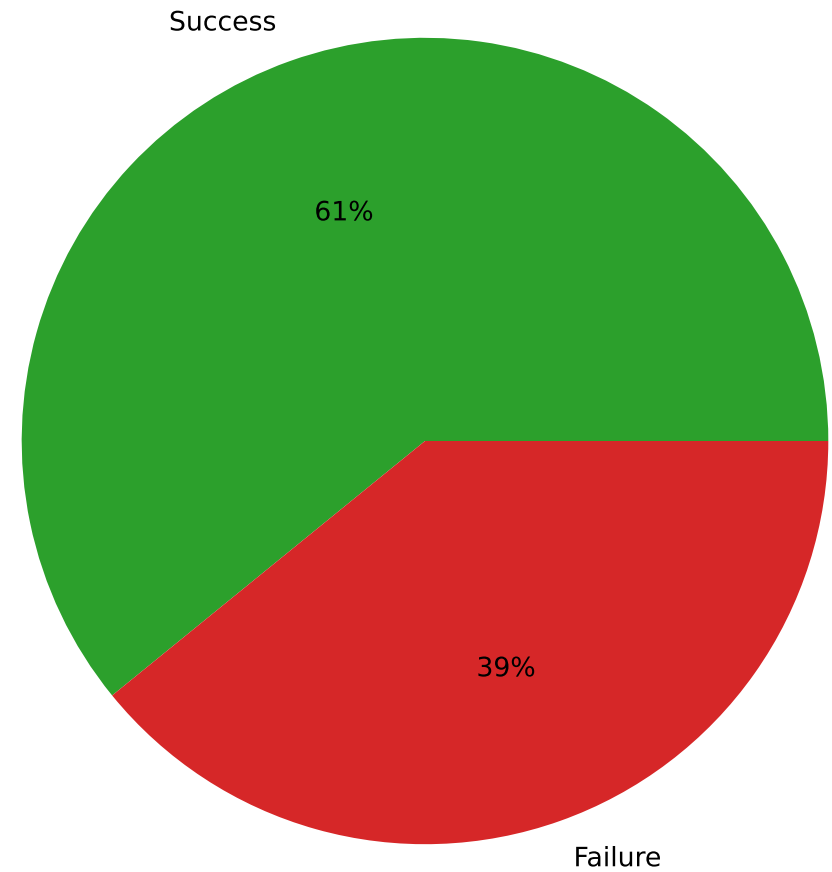
Distances are illustrative placeholders; the Folium script computes them with the haversine formula.

Plotly Dash: Success Pie Charts

All sites: share of total successes

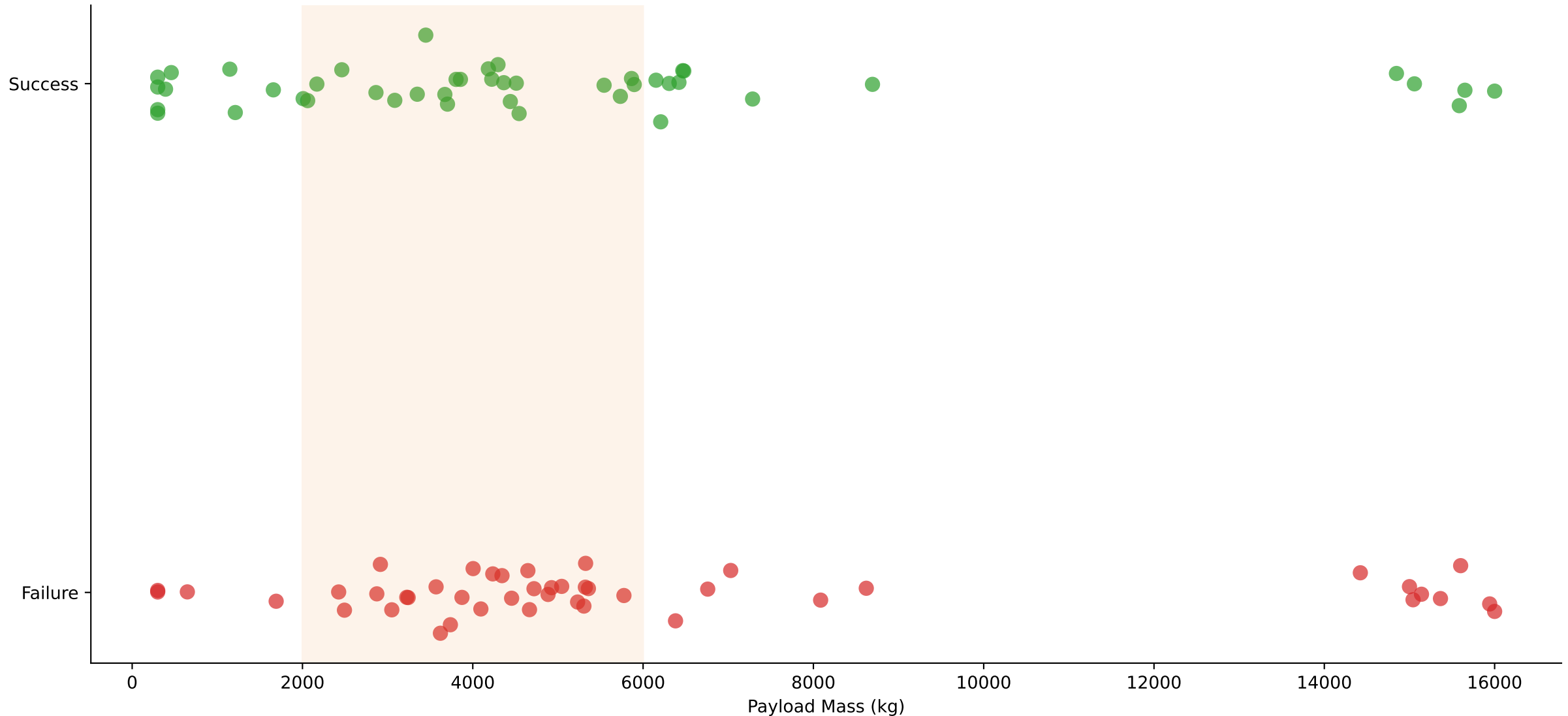


Highest success-ratio site: KSC LC-39A

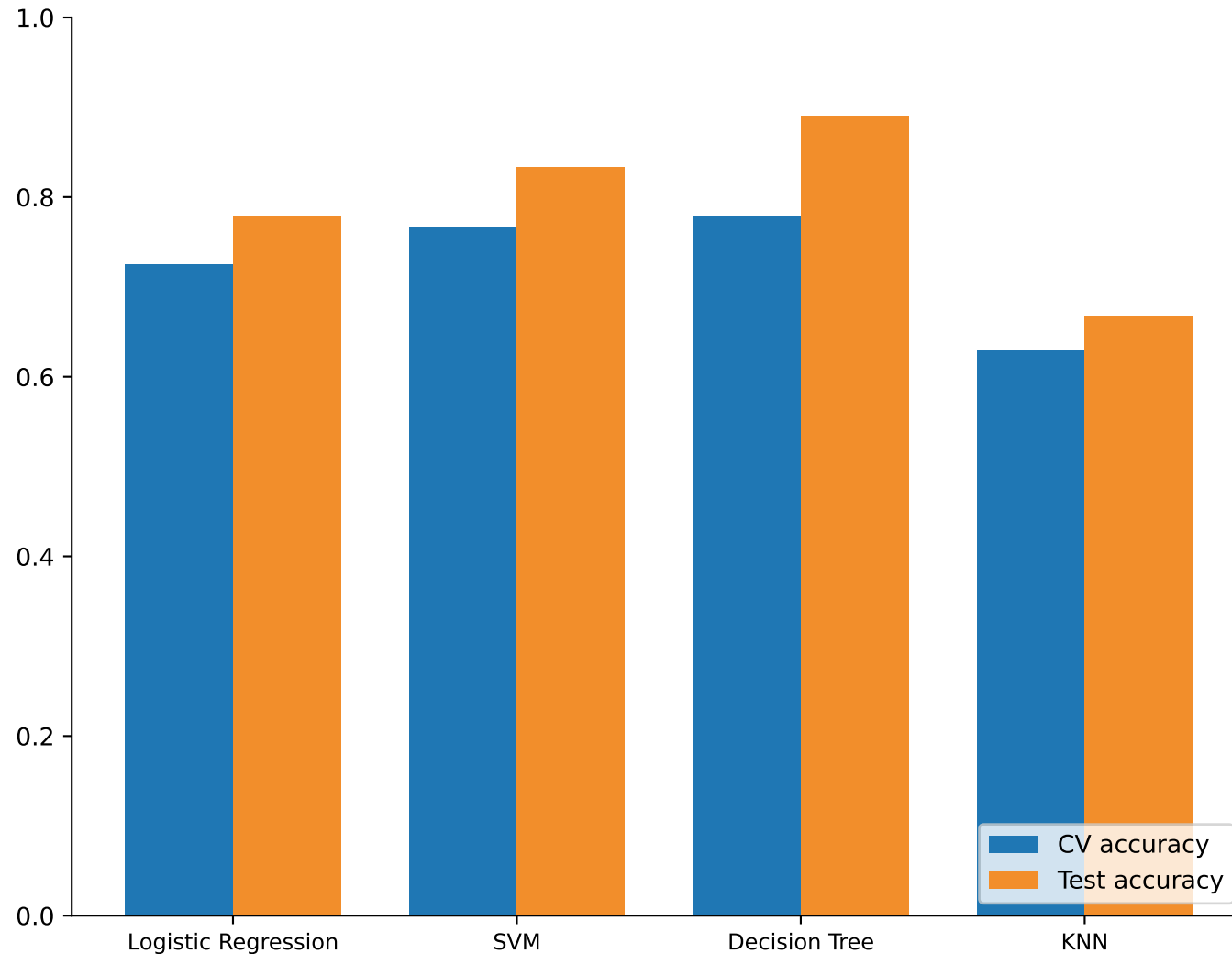


Plotly Dash: Payload vs Launch Outcome

RangeSlider 2,000-6,000 kg (shaded): success 45% vs 50% overall - compare with the full range via the slider.



Predictive Analysis: Model Comparison

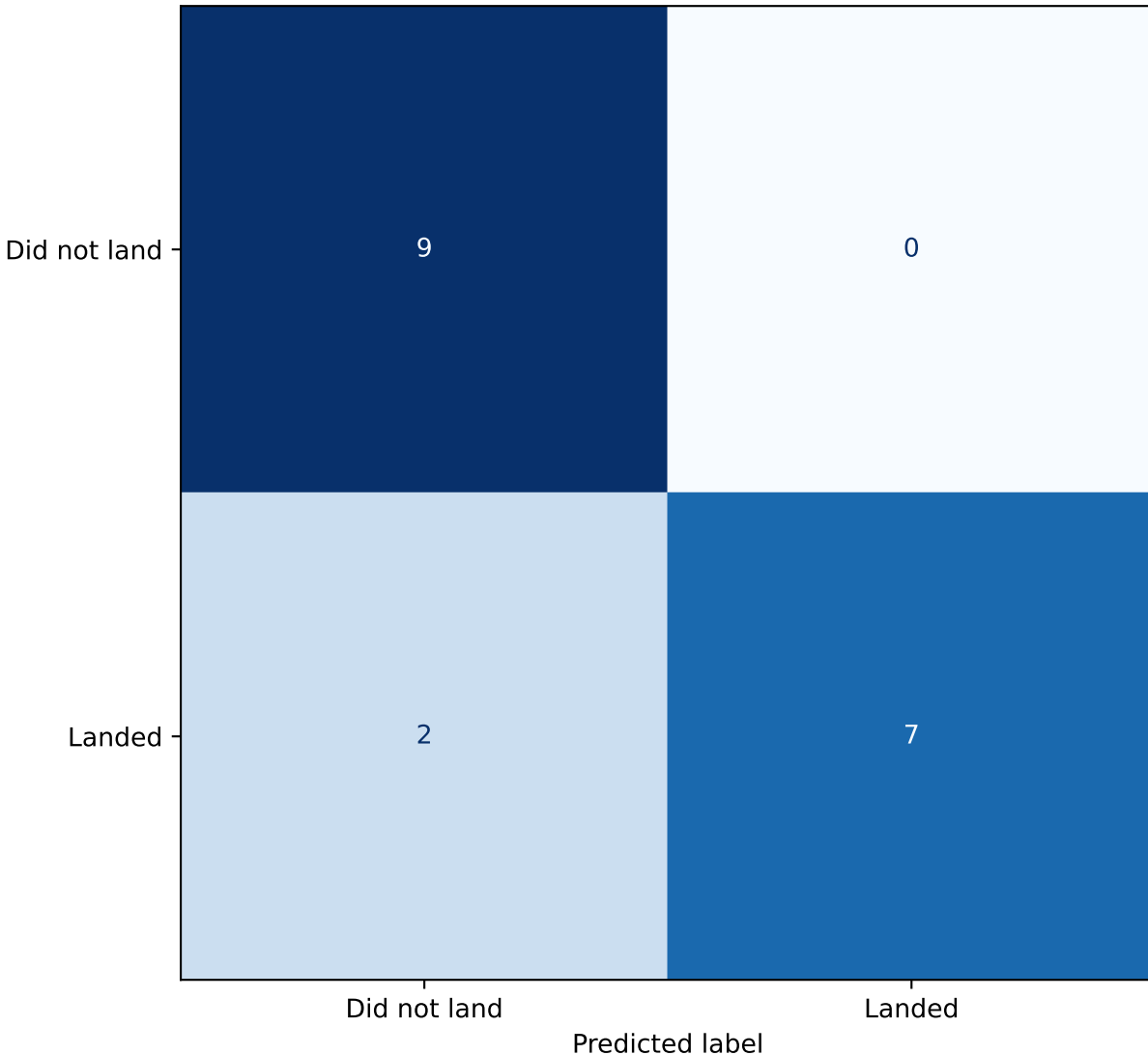


Results (10-fold GridSearchCV)

Model	CV	Test
Logistic Regression	0.72	0.78
SVM	0.77	0.83
Decision Tree	0.78	0.89
KNN	0.63	0.67

Best params (Decision Tree):
{ 'criterion': 'gini', 'max_depth': 2 }

Predictive Analysis: Confusion Matrix and Best Model



- Best model: Decision Tree (test acc 89%).
- Test set is only 18 launches, so differences between models are within noise.
- Errors: 0 false positive(s), 2 false negative(s).
- Flight number is the main signal in the data.
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- Next step: more data + cross-validated
- model comparison with confidence intervals.

Conclusion and Insights

- Landing success grew from 7% to 93% as flights accumulated: experience drives reliability.
- Orbit and payload effects are visible but noisy at this sample size (see orbit bar chart).
- Decision Tree reaches 89% test accuracy; all four models give a similar level of accuracy.
- Creative insight: pair the classifier with cost: $\text{expected cost} = \$165\text{M} - P(\text{land}) \times \sim \100M saving ,
- so a rival can price bids per mission profile.
- Insight: a 'learning-curve' feature (flights flown by the core version) would likely beat raw flight number.
- Future work: add weather, booster block and landing-pad data; try gradient boosting; deploy the Dash app.